CHASR: Causal Hallucination Attribution for Extreme Super-Resolution

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Abstract

Extreme super-resolution (xSR, $8\times/16\times$ and beyond) is severely ill-posed: the diffusion priors enabling plausible upscaling also make hallucinated detail unavoidable. Existing work measures how much a super-resolved image hallucinates but not why, limiting its use for forensic and task-driven applications where trust matters. We introduce CHASR, a causal attribution module decomposing hallucination risk at the pixel level into four causes: inherent ill-posedness, over-reliance on the generative prior, degradation-model mismatch, and distribution shift from the backbone's training data. CHASR wraps a frozen chained diffusion backbone with four lightweight signal estimators and a small trained fusion head producing a per-pixel five-way cause classification and a calibrated reliability score, evaluated on xSR-CausalBench, a synthetic benchmark with weak per-pixel ground truth. A first end-to-end validation at $16\times$ on real photographs shows the fusion head reaches 48.4–48.5% five-way pixel attribution accuracy, only 0.7–0.8 points above the benchmark's majority-class baseline, a margin a paired bootstrap does not distinguish from zero at 95% confidence. A capacity-matched logistic-regression control collapses to the majority-class rule, indicating the fusion head's small edge depends on its convolutional capacity rather than joint fitting alone. Recovery is highly uneven across causes: two region-local causes reach over 99% accuracy in their best examples, while degradation-mismatch is never recovered. A disentanglement analysis confirms

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an anticipated correlation between the null-space-uncertainty and prior-reliance signals. We report this pilot honestly, including where the numbers are weak; the central contribution is the causal taxonomy and attribution framework, not fidelity leaderboard performance.

Keywords: extreme super-resolution, hallucination attribution, diffusion models, causal analysis, null-space uncertainty, image forensics

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1. Introduction

A security camera captures a license plate at a distance where the plate occupies perhaps a dozen pixels. A satellite image resolves a rooftop as a handful of gray values. A decades-old photograph, scanned at low resolution, is the only surviving record of a document. In each case, someone eventually asks a diffusion-based super-resolution system to recover detail that the sensor never captured, and in each case the system will comply: modern extreme super-resolution (xSR) methods, operating at $8\times$ or $16\times$ upscaling factors, now produce images that look convincingly sharp. The trouble is that "looks convincingly sharp" and "is actually correct" have become only loosely related properties. At these factors the information-theoretic content of the input is a small fraction of what the output claims to show, and the generative prior inside the model is doing most of the work of deciding what the missing pixels should be. When it decides wrong, the result is hallucination: detail that is plausible, consistent with the model's training distribution, and false.

The severity of this problem scales directly with how the output will be used. A hallucinated license-plate character that happens to look legible is not a cosmetic defect if that character becomes evidence. A hallucinated texture on a satellite rooftop is not a rendering artifact if an analyst reports it as a structural feature. This is precisely the concern that motivates growing interest in extreme super-resolution's forensic and task-driven applications, and it explains why the special issue this paper targets asks explicitly for evaluation metrics that go beyond PSNR and SSIM: a fidelity score computed against a ground-truth image the deployment scenario does not have access to cannot tell an analyst whether the ten pixels they are looking at right now can be trusted.

A handful of recent papers move past fidelity to measure hallucination directly. HalluGen [1] introduces a controllable benchmark that separates intrinsic hallucination (violations of data consistency) from extrinsic hallucination (wrong content filled into the model's null space) and scores restorations against this two-way taxonomy. The Hallucination Score [2] takes a complementary approach, using a multimodal large language model to produce a single scalar severity score that can double as a differentiable fine-tuning signal. Both are genuine advances, and both share a structural limitation for the use cases this paper is concerned with: they tell a user *how much* a region hallucinates, or at best which of two broad categories the hallucination falls into, but not *why*. An analyst deciding whether to trust a specific license-plate character needs more than a severity number. They need to know whether the character is uncertain because the optics genuinely destroyed that information (in which case no model, however good, could have recovered it correctly), because the model ignored usable evidence in favor of its own prior (a model defect, potentially fixable), because the model was applied to a degradation it was never trained to handle (an operating-conditions mismatch), or because the content itself is unlike anything the model has seen (an out-of-distribution failure). These are four distinguishable failure modes with four different practical responses, and no existing method attributes hallucination to a specific one of them.

This paper introduces CHASR (Causal Hallucination Attribution for Super-Resolution), a module that sits on top of a frozen extreme super-resolution backbone and produces, for every pixel of the output, both a reliability score and a classification into one of four causes or a fifth "reliable" class. The design commits to four causal signals that map directly onto the four failure modes above: null-space uncertainty, measured as the pixelwise variance across multiple degradation-consistent stochastic samples from the backbone; prior-reliance, measured as the relative sensitivity of the output to the low-resolution evidence versus to the model's random seed; degradation-model mismatch, measured by re-estimating the input's blur and noise parameters with an independent blind kernel estimator and comparing them against the backbone's assumed training distribution; and distribution shift, measured as the nearest-neighbor distance from the output's frozen self-supervised features to a bank of the backbone's training-distribution features. A small fusion head, the only trained component in the entire pipeline, learns to combine these four raw signals with a shallow image feature into a calibrated per-pixel attribution.

Training and evaluating this fusion head requires per-pixel ground truth for which cause produced a given hallucination, and no such labels exist for real images because the true cause of a real hallucination is, definitionally, unobservable. We address this by constructing xSR-CausalBench, a synthetic benchmark that extends HalluGen's controllable-hallucination construction from two causes to four: four distinct procedures inject ill-posedness, prior-reliance, degradation-mismatch, or distribution-shift dominance into an otherwise normal image under experimentally controlled conditions, producing a weak per-pixel label mask alongside each synthesized low-resolution and high-resolution pair.

The causes this taxonomy separates are not independent in practice, and pretending otherwise would misrepresent what the framework can promise. A region with extreme ill-posedness naturally invites more prior reliance, because there is less real evidence available for the model to weigh against its prior in the first place. Section 5 reports this correlation directly, measured from the pipeline's own output rather than assumed, because the alternative, quietly hoping the four signals turn out to be independent and finding out otherwise at review time, serves nobody.

This paper's contribution is fourfold: a four-way causal taxonomy for extreme super-resolution hallucination, extending the two-way intrinsic/extrinsic split in the closest prior work; the CHASR architecture that instantiates the taxonomy as four computable signals plus a learned fusion head, built on a publicly reproducible diffusion backbone; xSR-CausalBench, a synthetic benchmark that generates per-pixel weak ground truth for all four causes; and an honestly reported pilot validation, at real 16× upscaling on natural photographs, that demonstrates the pipeline is trainable and produces above-chance, structurally sensible attribution, while being explicit about what remains to scale before the numbers reported here should be read as a final answer rather than a first one.

2. Related Work

2.1. Blind and diffusion-based super-resolution

Classical blind super-resolution treats the degradation kernel as an unknown to be estimated jointly with, or prior to, the restoration itself. KernelGAN [3] estimates a per-image blur kernel using an internal GAN trained purely on patch recurrence within the low-resolution image, requiring no external training data. MANet [4] instead learns a mutual affine network that estimates spatially variant kernels directly, relaxing the common assumption of a single global kernel per image. DASR [5] sidesteps explicit kernel parameterization entirely, learning an unsupervised degradation representation via contrastive learning and conditioning the restoration network on it directly. This lineage of blind kernel estimation is a component CHASR reuses directly: Signal 3 (degradation-model mismatch) is computed by an independent blind estimator trained on both the backbone's assumed degradation pool and an out-of-distribution pool, then compared against the backbone's training assumptions, rather than proposed here as a new estimation method.

Diffusion models have displaced GAN-based approaches as the dominant paradigm for photorealistic super-resolution at moderate factors. StableSR [6] adapts a pretrained text-to-image diffusion prior for real-world restoration through a time-aware encoder and a controllable feature-wrapping module. SeeSR [7] conditions the diffusion process on semantic tags extracted from the degraded input, improving fidelity to the true image content rather than merely to a plausible one. PASD [8] introduces pixel-aware cross-attention so that the diffusion process attends explicitly to which regions the input actually constrains. Chain-of-Zoom [9] pushes this diffusion-prior paradigm to substantially more extreme factors by reformulating extreme super-resolution as an autoregressive sequence of moderate zoom-in steps guided by vision-language model text prompts, each step handled by an off-the-shelf 4× diffusion model rather than a single model retrained for the full factor. CHASR's backbone follows this same chained-hop principle for reaching 16× from a 4× checkpoint, though for reproducibility on modest hardware it wraps the publicly downloadable `stable-diffusion-x4-upscaler` checkpoint directly rather than a VLM-guided variant, trading some of Chain-of-Zoom's content-awareness for a fully open, easily reproduced pipeline. Real-ESRGAN [10] remains the standard GAN-based reference point for real-world super-resolution trained entirely on synthetic degradations, and Section 4 reports it as a real fidelity baseline, chained to the same 16× factor as CHASR's own backbone, rather than only citing it here. Two concurrent lines of work couple diffusion-based super-resolution directly to an uncertainty signal rather than treating uncertainty as an afterthought: UGDiff [11] estimates per-latent-feature reconstruction uncertainty and uses it to steer high-frequency detail restoration toward high-uncertainty regions while leaving confident regions untouched, and QUSR [12] adapts injected diffusion noise strength to local uncertainty alongside a multimodal-LLM-based quality prior, under unknown, spatially varying degradation. Both use uncertainty as a single scalar steering signal for where to generate more aggressively; CHASR's Signal 1 measures a structurally similar null-space quantity but treats it as one of four distinguishable causes to attribute after generation rather than a knob to steer generation with, a diagnostic rather than a generative use of the same underlying idea.

2.2. Measuring hallucination in generative restoration

HalluGen [1] is the closest prior work to CHASR's evaluation methodology. It separates hallucination into intrinsic (violating data consistency with the observed degradation) and extrinsic (wrong content filled into the null space where data consistency alone does not constrain the answer) categories, constructs a synthetic controllable-hallucination benchmark via diffusion posterior sampling, and proposes a patch-level severity metric. Its own benchmark is built from low-field MRI restoration rather than natural photographs, so CHASR's xSR-CausalBench (Section 3.5) is a cross-domain methodological descendant rather than a direct reuse: it ports the controllable-injection idea from medical to natural-image extreme super-resolution, extending it from two categories to four and from a severity score to a per-pixel causal label. The Hallucination Score [2] takes a different but complementary route: rather than decomposing hallucination by cause, it uses a multimodal large language model to judge two properties, implausibility under any degradation and semantic deviation from what the input evidence supports, and combines them into a scalar that doubles as a reward signal for fine-tuning. Both works measure severity or coarse category; neither attributes a specific hallucinated region to one of several distinguishable causes, which is the gap this paper targets directly. The same detection-versus-attribution gap has emerged concurrently outside super-resolution: Vo et al. [13] decompose vision-language model failures on knowledge-intensive visual question answering into recognition-stage versus knowledge-stage causes rather than a single failure score, enabling different fixes (image repair, entity support, question rewriting) for different causes, the same why-before-what argument this paper makes for extreme super-resolution. In a related line of the present author's own work, addressing a structurally similar problem in a different modality, Dang et al. [14] intercept and correct hallucinated tokens during vision-language model generation via a dynamically constructed, externally-verified knowledge graph rather than a post-hoc severity score, underscoring that mechanism-aware, causal intervention is an emerging theme across generative-AI hallucination research generally, not a concern specific to image restoration.

At a more general level, work on diffusion models outside restoration has begun explaining hallucination as an out-of-distribution phenomenon: Aithal et al. [15] show that diffusion models can hallucinate through mode interpolation, generating samples in regions between distinct modes of the training distribution that the model never actually observed, which is a mechanistic account of why unfamiliar content produces confident-looking but wrong output. Signal 4 in this paper's method operationalizes the restoration-specific analogue of that same intuition, measuring an output's feature-space distance from the backbone's training distribution directly, rather than proposing a new theory of why distribution shift causes hallucination.

Range-null space decomposition, formalized for diffusion-based zero-shot inverse problems by DDNM [16], provides the mathematical grounding for CHASR's Signal 1. DDNM shows that for a known linear degradation operator, a diffusion sampling trajectory can be projected at every step so that the range-space component exactly matches the observation while the diffusion prior is left free to fill the null space, and demonstrates this on a range of restoration tasks under the assumption that the degradation operator is known in advance. CHASR uses the same range-null projection, applied to the average-pooling operator that governs its synthetic degradation pipeline, to guarantee that every stochastic sample the backbone produces is exactly degradation-consistent, which is what makes the sample-to-sample variance in Signal 1 attributable to null-space uncertainty specifically rather than to a data-consistency violation. Where DDNM assumes a known degradation and does not address attribution, CHASR assumes a partially unknown degradation (estimated blindly for Signal 3) and uses the resulting variance as one input to a diagnostic rather than as the whole restoration method.

2.3. Task-driven super-resolution and this special issue

The special issue this paper targets is organized in part around task-driven reliability, and closely related work on embedding-similarity-guided license-plate super-resolution [17] exemplifies the concern directly: a super-resolved license plate is only useful if the specific characters it renders are correct, not merely if the overall image looks sharp. The concurrent XLPSR Grand Challenge at ICIP 2026, whose finalist entries include ensemble approaches combining super-resolution backbones with scene-text recognition voting [18], reflects the same practical stakes at competition scale: entries are scored on downstream recognition accuracy, not perceptual quality. CHASR's downstream case study, correlating flagged high-mismatch or high-distribution-shift regions with actual optical character recognition failure on text-patch crops rather than treating task-driven evaluation as secondary to fidelity, is implemented as a reusable module (Section 5) but not yet exercised against real recognition failures in this pilot.

3. Method

CHASR is a diagnostic module, not a new super-resolution architecture. It wraps a frozen backbone and adds four causal signal estimators plus a small trained fusion head, so that the entire training cost of the system is confined to the fusion head while the backbone's own super-resolution quality is left exactly as published. Figure 1 shows the full pipeline: the frozen backbone (Section 3.2) produces the super-resolved output and the raw material each signal estimator (Section 3.3) needs, and the only trained component, the fusion head (Section 3.4), combines the four resulting signal maps into a per-pixel cause classification and reliability score.

3.1. Problem formulation

Let $x \in [0,1]^{H \times W \times 3}$ denote a high-resolution image and A_r a linear degradation operator that block-average-pools by factor r, so for every non-overlapping $r \times r$ block B,

$$(A_r x)_B = \frac{1}{r^2} \sum_{i \in B} x_i. \tag{1}$$

The synthetic low-resolution observation is $y = A_r(k * x)$, where k is a blur kernel applied before pooling. Every blur kernel used in this paper, whether it produces an in-distribution or out-of-distribution degradation, is drawn from the same general anisotropic Gaussian family,

$$k(u, v\,;\, \sigma_x, \sigma_y, \theta) \propto \exp\left(-\frac{u_\theta^2}{2\sigma_x^2} - \frac{v_\theta^2}{2\sigma_y^2}\right), \quad \begin{bmatrix} u_\theta \\ v_\theta \end{bmatrix} = \begin{bmatrix} \cos\theta & \sin\theta \\ -\sin\theta & \cos\theta \end{bmatrix} \begin{bmatrix} u \\ v \end{bmatrix}, \tag{2}$$

normalized to sum to 1 over its support; the standard (in-distribution) pool fixes $\sigma_x = \sigma_y$ and $\theta = 0$ (isotropic), while the mismatch pool draws σ_x, σ_y

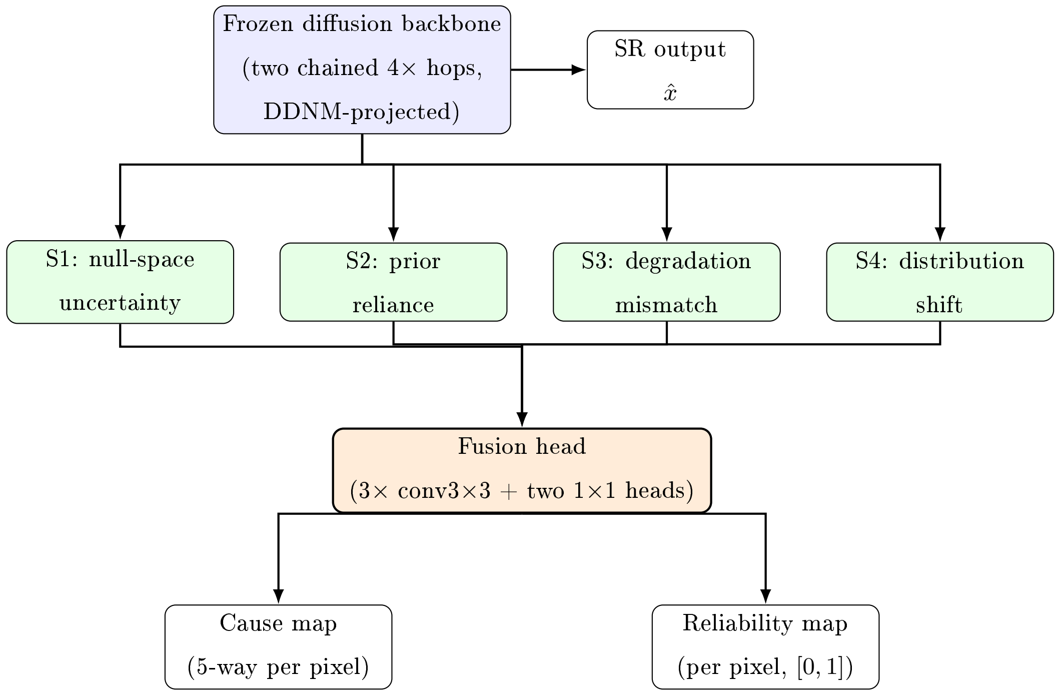

Figure 1: CHASR pipeline. Blue: frozen backbone. Green: the four independent signal estimators, each reading the backbone's output and/or intermediate state (Section 3.3). Orange: the fusion head, the only trained component, which stacks the four signal maps with a shallow luminance feature and predicts a per-pixel cause and reliability.

independently from a disjoint, larger range and θ from a nonzero range (Section 3.5), making Equation 2 the shared parametric family Signal 3's estimator regresses $(\hat{\sigma}_x, \hat{\sigma}_y, \hat{\theta})$ against. For simplicity, $y = A_r(k * x)$ omits two further steps the actual degradation pipeline applies after blur and pooling, additive Gaussian noise and JPEG recompression at a randomly sampled quality, both drawn from the same standard/mismatch pool distinction as k itself; Signal 3's estimator additionally regresses the noise level, so its full input is the parameters of this richer, three-step pipeline rather than the blur kernel alone.

Extreme super-resolution seeks $\hat{x} \approx x$ from y alone at $r = 16$, a severely underdetermined inverse problem: A_{16} discards 255/256 of the pixel-level information in x, so any $\hat{x}$ satisfying $A_{16}(\hat{x}) = y$ is a valid data-consistent completion regardless of whether it matches the true x. CHASR does not attempt to resolve this ill-posedness; it attributes it. Given the frozen backbone's output $\hat{x}$ and a fixed taxonomy of causes $\mathcal{C} = \{\texttt{ILL\_POSED}, \texttt{PRIOR\_RELIANCE}, \texttt{MISMATCH}, \texttt{DIST\_SHIFT}, \texttt{RELIABLE}\}$, the task this paper addresses is learning a per-pixel classifier

$$\hat{c}(p) = \arg\max_{c \in \mathcal{C}} \; f_\theta(S_1(p), S_2(p), S_3(p), S_4(p), L(p))_c \,, \tag{3}$$

where $S_1, \ldots, S_4$ are the four causal signals defined in Section 3.3, L is a shallow luminance feature, and f_θ is the fusion head of Section 3.4 — the only component in Equation 3 with learned parameters θ; every signal on its right-hand side is computed from a frozen or independently pretrained model.

3.2. Frozen backbone

The backbone reaches 16× from 64×64 low-resolution patches through two chained 4× diffusion hops, using the publicly downloadable checkpoint `stable-diffusion-x4-upscaler` (hosted on the Hugging Face Hub under the `stabilityai` organization) at each hop for full reproducibility on a single consumer GPU. The first hop upscales the full 64×64 patch to 256×256. The second hop is applied per-tile, splitting the 256×256 intermediate into four 128×128 tiles, each independently upscaled to 512×512 and stitched back into the final 1024×1024 output, which keeps peak memory within an 8 GB budget at the cost of a faint seam artifact at tile boundaries that does not affect the causal signals computed here. After every hop, the output is projected via the DDNM range-null decomposition [16] against the true low-resolution observation at that hop's scale. For a linear degradation operator A_r, DDNM's general projection replaces a raw diffusion estimate $\hat{x}_\theta$ with

$$\hat{x}' = A_r^\dagger y + \left(I - A_r^\dagger A_r\right) \hat{x}_\theta, \tag{4}$$

where $A_r^\dagger$ is the Moore–Penrose pseudo-inverse of A_r: the first term fixes the range space to exactly match the observation, and the second leaves the null space, everything A_r cannot see, free for the diffusion prior to fill. For the block-average-pooling operator of Equation 1, A_r has orthogonal rows ($A_r A_r^\top = r^{-2} I$), so $A_r^\dagger y$ broadcasts each observed value y_B uniformly across its source block, and $A_r^\dagger A_r \hat{x}_\theta$ replaces every pixel with its own block mean. Equation 4 then reduces, block by block, to

$$\hat{x}_i' = \hat{x}_{\theta,i} - \frac{1}{r^2}\sum_{j\in B}\hat{x}_{\theta,j} \;+\; y_B, \qquad \forall\, i \in B, \tag{5}$$

the closed-form, per-block correction actually implemented: each pixel is shifted by the difference between its block's true observed mean and the diffusion output's own block mean. This guarantees every sample the backbone returns exactly satisfies $A_r(\hat{x}') = y$, at every hop, which is what makes the variance across independently sampled outputs attributable specifically to null-space content rather than to the model occasionally violating the observation. Algorithm 1 states the full chained procedure, applying Equation 5 at each hop's own scale.

Algorithm 1: Chained, DDNM-projected 16× sampling

Input : Low-resolution patch $y \in \mathbb{R}^{64\times 64\times 3}$, sample count K

Output: K degradation-consistent 16× super-resolved samples $\{\hat{x}_1, \ldots, \hat{x}_K\}$

for $k \leftarrow 1$ **to** K **do**

  $z_1 \leftarrow \mathrm{DiffusionHop}_1(y, \mathrm{seed} = k)$; `// ` $64\times 64 \to 256\times 256$

  $z_1 \leftarrow \mathrm{DDNM\_project}(z_1, y, \mathrm{scale} = 4)$; `// exact consistency at hop 1's scale`

  for tile $\in \mathrm{split}(z_1, 4)$ **do**

    $t \leftarrow \mathrm{DiffusionHop}_2(\mathrm{tile}, \mathrm{seed} = k)$; `// ` $128\times 128 \to 512\times 512$

    $t \leftarrow \mathrm{DDNM\_project}(t, \mathrm{tile}, \mathrm{scale} = 4)$;

    stitch(t) into $z_2 \in \mathbb{R}^{1024\times 1024\times 3}$;

  end

  $\hat{x}_k \leftarrow \mathrm{DDNM\_project}(z_2, y, \mathrm{scale} = 16)$; `// final consistency check against` y `directly`

end

return $\{\hat{x}_1, \ldots, \hat{x}_K\}$

3.3. Four causal signals

Signal 1: null-space uncertainty. The backbone is sampled K times with different random seeds, each sample $\hat{x}_1, \ldots, \hat{x}_K$ independently satisfying Equation 5. The pixelwise, channel-averaged unbiased sample variance across the K samples,

$$\bar{S}_1(p) = \frac{1}{3} \sum_{ch=1}^{3} \frac{1}{K-1} \sum_{k=1}^{K} \left(\hat{x}_k(p, ch) - \bar{x}(p, ch)\right)^2, \qquad \bar{x}(p, ch) = \frac{1}{K} \sum_{k=1}^{K} \hat{x}_k(p, ch), \tag{6}$$

is then rescaled to $[0, 1]$ against the theoretical maximum unbiased two-sample variance a $[0, 1]$-bounded pixel value can attain (achieved when $\hat{x}_1(p) = 0, \hat{x}_2(p) = 1$, giving $\bar{S}_1^{\max} = 0.5$),

$$S_1(p) = \min\!\left(1,\ \bar{S}_1(p) \,/\, \bar{S}_1^{\max}\right). \tag{7}$$

Because data consistency is fixed by construction, the samples can only differ in their null-space content, so high variance at a pixel means the ill-posedness at that pixel is large enough that the model genuinely could not have determined a unique answer from the evidence, independent of whether the model itself is well-calibrated.

Signal 2: generative-prior over-reliance. This signal asks a different question: even where evidence is available, did the model use it? For the frozen diffusion backbone, whose stochasticity is a discrete random seed rather than a continuous latent, this is estimated by finite differences at the first (256×256) hop: the raw, pre-projection output's sensitivity to a small perturbation of the low-resolution evidence (g_{evidence}) is compared against its sensitivity to a full reseed at fixed evidence (g_{prior}),

$$S_2(p) = \frac{g_{\text{prior}}(p)}{g_{\text{prior}}(p) + g_{\text{evidence}}(p) + \varepsilon}. \tag{8}$$

Both terms are compared as raw output-magnitude changes on the same units, rather than as normalized derivatives, because normalizing only the continuous perturbation by its step size while leaving the discrete reseed unnormalized would bias the ratio toward zero regardless of the model's actual behavior.

A general finite-difference formulation for a continuously perturbable forward function and latent is also implemented as a standalone, independently tested utility for future backbones with a true continuous latent, but is not used in the real pipeline for exactly this reason.

Signal 3: degradation-model mismatch. An independent, lightweight convolutional network is trained to regress a blur kernel's spatial parameters (anisotropic sigma $(\hat{\sigma}_x, \hat{\sigma}_y)$, rotation angle $\hat{\theta}$) and additive noise level directly from a 64×64 patch, trained on a 50/50 mixture of the backbone's assumed in-distribution degradation pool (upper sigma bound $\sigma_{\max}$, zero rotation) and an out-of-distribution pool with a disjoint, larger sigma range and non-zero rotation. At inference, this estimator re-examines the actual input patch and scores how far the estimate falls outside the assumed pool,

$$S_3 = \min\left(1,\ \frac{\max(0,\, \max(\hat{\sigma}_x, \hat{\sigma}_y) - \sigma_{\max}) + |\hat{\theta}|}{d_{\max}}\right), \tag{9}$$

normalized by the out-of-distribution pool's own worst-case raw distance $d_{\max}$ so the signal saturates at 1.0 only at genuinely extreme mismatch rather than prematurely. S_3 is a single scalar per input patch, broadcast uniformly to every pixel, since the degradation kernel is assumed spatially constant within one patch. A high value here means the backbone is being asked to restore an image degraded in a way it was never exposed to during training, independent of whether the content itself is familiar.

Signal 4: distribution shift. A frozen, self-supervised DINOv2 encoder [19] extracts a feature vector f from the super-resolved output, and its k-nearest-neighbor distance $d_{\mathrm{kNN}}(f, \mathcal{B})$ to a reference bank $\mathcal{B}$ of features extracted once from the backbone's training-distribution images is computed. Because a sparse reference bank in a high-dimensional feature space suffers a curse-of-dimensionality effect at $k > 1$, where a near-duplicate query's second through k-th neighbors are effectively generic points at a roughly constant distance that washes out fine-grained separation, the raw distance is rescaled by the bank's own internal typical k-nearest-neighbor distance $s(\mathcal{B})$ among its own members,

$$S_4 = \frac{d_{\mathrm{kNN}}(f, \mathcal{B})}{d_{\mathrm{kNN}}(f, \mathcal{B}) + s(\mathcal{B})}, \tag{10}$$

producing a self-calibrated score in $[0, 1)$ without assuming a fixed feature-space magnitude in advance. A high value here means the output's own semantic content sits far outside the region of feature space the backbone was actually trained on, relative to how tightly clustered the training distribution's own features are, independent of whether the degradation kernel or the ill-posedness level were themselves anomalous. Like S_3, S_4 is computed once per output image and broadcast to every pixel.

Algorithm 2 collects the four signal computations (Equations 6–10) in one place, making explicit which quantities each signal shares (all four read from the same K-sample set of Algorithm 1, and S_2 additionally requires two extra single-hop calls beyond it) and which are independently pretrained rather than computed fresh per pixel (S_3's kernel estimator E_θ and S_4's DINOv2 encoder ϕ are both frozen, queried here, not trained here).

3.4. Fusion head

The four raw signals, each independently rescaled to a comparable $[0, 1]$-ish range before this stage, are stacked with one shallow image-luminance feature into a five-channel input $\mathbf{S}(p) = [S_1(p), S_2(p), S_3(p), S_4(p), L(p)]$ and passed through a small convolutional network f_θ: three 3×3 convolutional layers followed by two separate 1×1 heads, one producing a five-way per-pixel classification logit $f_\theta^{\mathrm{cause}}(\mathbf{S})(p) \in \mathbb{R}^5$ over $\mathcal{C}$ (four causes plus a reliable class), the other a sigmoid-activated reliability score $f_\theta^{\mathrm{rel}}(\mathbf{S})(p) \in [0, 1]$. Writing $y(p) \in \mathcal{C}$ for xSR-CausalBench's ground-truth label at pixel p and Ω for the set of all labeled training pixels, both heads are trained jointly by minimizing

$$\mathcal{L}(\theta) = \frac{1}{|\Omega|} \sum_{p \in \Omega} \Big[\underbrace{-\log \mathrm{softmax}\big(f_\theta^{\mathrm{cause}}(\mathbf{S})(p)\big)_{y(p)}}_{\text{cause cross-entropy}} + \underbrace{\mathrm{BCE}\Big(f_\theta^{\mathrm{rel}}(\mathbf{S})(p),\, \mathbb{1}\big[y(p) = \texttt{RELIABLE}\big]\Big)}_{\text{reliability binary cross-entropy}} \Big], \tag{11}$$

Algorithm 2: Four-signal computation

Input : LR patch y, samples $\{\hat{x}_1, \ldots, \hat{x}_K\}$ from Algorithm 1, kernel estimator E_θ, DINOv2 encoder ϕ, reference bank $\mathcal{B}$

Output: Signal stack $[S_1, S_2, S_3, S_4] \in [0,1]^{4\times1024\times1024}$

$S_1 \leftarrow$ Eq. 6, then normalized via Eq. 7;

$h_0 \leftarrow \mathrm{DiffusionHop}_1(y, \mathrm{seed} = 0)$;

$h_{\mathrm{ev}} \leftarrow \mathrm{DiffusionHop}_1(y + \epsilon \cdot \mathcal{N}(0,1), \mathrm{seed} = 0)$; `// evidence perturbation, fixed seed`

$h_{\mathrm{pr}} \leftarrow \mathrm{DiffusionHop}_1(y, \mathrm{seed} = 1)$; `// prior perturbation, fixed evidence`

$g_{\mathrm{evidence}} \leftarrow |h_{\mathrm{ev}} - h_0|$, $g_{\mathrm{prior}} \leftarrow |h_{\mathrm{pr}} - h_0|$;

$S_2 \leftarrow$ Eq. 8, upsampled to 1024×1024;

$(\hat{\sigma}_x, \hat{\sigma}_y, \hat{\theta}, \cdot) \leftarrow E_\theta(y)$;

$S_3 \leftarrow$ Eq. 9, broadcast to all pixels;

$f \leftarrow \phi(\hat{x}_1)$; `// DINOv2 features of the first sample`

$S_4 \leftarrow$ Eq. 10, broadcast to all pixels;

return $[S_1, S_2, S_3, S_4]$

with $\hat{c}(p)$ of Equation 3 recovered at inference as $\arg\max_c \mathrm{softmax}(f_\theta^{\mathrm{cause}}(\mathbf{S})(p))_c$. This is the only trained component in the entire pipeline; the backbone, the kernel estimator, and the DINOv2 encoder are all frozen or independently pre-trained and contribute no term to Equation 11.

3.5. xSR-CausalBench

Training and evaluating the fusion head requires per-pixel labels for which cause dominates at each pixel, and because the true cause of a hallucination in a real photograph is unobservable, these labels must come from controlled synthetic construction rather than from annotation. Building on HalluGen's diffusion-posterior-sampling controllable-hallucination idea [1], four procedures each take a held-out high-resolution source image and produce a low-resolution and high-resolution pair with a weak per-pixel label mask over five classes (the four causes plus reliable):

Ill-posedness-dominant degrades the entire image with the backbone's standard, correctly specified degradation model at the full 16× factor, labeling the whole image ill-posed by construction, since extreme downsampling alone is sufficient to destroy most of the null-space content regardless of anything else.

Prior-reliance-dominant degrades the image with the same standard model, but first zeroes out the evidence in a rectangular sub-region covering 30% of the image's height and width (9% of its area), placed at a uniformly random valid offset, beyond what the nominal degradation alone would remove; the exact zeroed rectangle, not a fixed or scale-rounded approximation of it, is labeled prior-reliance-dominant, and the remainder reliable.

Degradation-mismatch-dominant degrades the entire image with an out-of-training-distribution kernel, anisotropic and rotated with a sigma range disjoint from and above the standard pool, while content stays in-distribution, labeling the whole image degradation-mismatch-dominant.

Distribution-shift-dominant blends a small patch, 64×64 pixels (0.4% of the 1024×1024 image area), into an otherwise normal image before applying the standard degradation model, then labels exactly the blended region distribution-shift-dominant and the remainder reliable. The patch content is drawn from a pool of natural images held disjoint from every procedure's source images (never from the same image being degraded), rather than from any exotic or synthetically out-of-domain texture source; this is a proxy for distribution shift using ordinary but contextually foreign content, not a claim that the patch content is itself unusual in any absolute sense, and Section 5 treats the strength of this proxy as an open question.

Each source image, drawn from a held-out validation split, is passed through all four procedures independently and deterministically per index, so the benchmark is reproducible given a fixed seed. Two of the four procedures assign a label from the physical operation applied (ill-posedness and degradation-mismatch, both whole-image), while the other two assign a label from where evidence was deliberately altered relative to the standard degradation (prior-reliance and distribution-shift, both region-local); this means the same standard degradation operator underlies the ill-posedness procedure's whole-image ill-posed label and the prior-reliance procedure's remainder-region reliable label, with the two procedures differentiated only by whether the sub-region evidence was additionally destroyed before that shared degradation was applied. This is a real construct-validity limitation, not a subtlety hidden from the reader: the ground-truth label reflects which procedure generated the pixel, not an independently verified causal judgment about that pixel in isolation, and Section 5 returns to it directly. These four causes were not expected to separate perfectly; Section 5 reports the resulting entanglement directly. Algorithm 3 states the four-procedure dispatch precisely enough to reimplement without reading the source: given a fixed per-item seed, exactly one of the four procedures runs, deterministically, so re-generating the benchmark from the same (image list, seed) pair reproduces every pixel of every label mask exactly.

Algorithm 3: xSR-CausalBench sample construction

Input : HR source image x, OOD patch pool $\mathcal{P}$, procedure
$c \in \{$ill_posed, prior_reliance, mismatch, distribution_shift$\}$,
seed s

Output: LR/HR pair (y, x'), per-pixel label map $L \in \{0, \ldots, 4\}^{H \times W}$

rng $\leftarrow$ Generator(s);
switch c **do**
    case *ill_posed* **do**
        $y \leftarrow$ degrade_standard$(x, 16, \mathrm{rng})$; $x' \leftarrow x$;
        $L \leftarrow$ ILL_POSED everywhere;
    end
    case *prior_reliance* **do**
        $(p_h, p_w) \leftarrow (0.3H, 0.3W)$; $(y_0, x_0) \leftarrow$ uniform random valid offset;
        $\tilde{x} \leftarrow x$ with region $[y_0{:}y_0{+}p_h,\ x_0{:}x_0{+}p_w]$ set to 0;
        $y \leftarrow$ degrade_standard$(\tilde{x}, 16, \mathrm{rng})$; $x' \leftarrow x$;
        $L \leftarrow$ RELIABLE everywhere, then PRIOR_RELIANCE in that region;
    end
    case *mismatch* **do**
        $y \leftarrow$ degrade_mismatch$(x, 16, \mathrm{rng})$; `// OOD kernel:` σ `above standard range,` $\theta \neq 0$
        $x' \leftarrow x$; $L \leftarrow$ DEGRADATION_MISMATCH everywhere;
    end
    case *distribution_shift* **do**
        $p \leftarrow \mathcal{P}[\text{image\_index} \bmod |\mathcal{P}|]$; `// 64×64 patch, round-robin`
        $(y_0, x_0) \leftarrow$ uniform random valid offset for p;
        $\tilde{x} \leftarrow x$ with region $[y_0{:}y_0{+}64,\ x_0{:}x_0{+}64]$ replaced by p;
        $y \leftarrow$ degrade_standard$(\tilde{x}, 16, \mathrm{rng})$; $x' \leftarrow \tilde{x}$;
        $L \leftarrow$ RELIABLE everywhere, then DISTRIBUTION_SHIFT in that region;
    end
end
return $(y, x'), L$

4. Experiments

4.1. Setup

All experiments use the 16× backbone described in Section 3.2, unmodified from its published checkpoint. The blind kernel estimator (Signal 3) was trained for 20 epochs on 500 images from the DIV2K training split [20], resized to 256×256 patches, with the 50/50 standard/mismatch degradation mixture described in Section 3.3. The DINOv2 reference feature bank (Signal 4) was built once from 800 DIV2K training images (a superset of the kernel estimator's 500), producing an 800-entry bank of 384-dimensional features. The fusion head was trained for 30 epochs on xSR-CausalBench instances built from 40 DIV2K validation images (160 training items across the four procedures, $K = 2$ stochastic backbone samples per item), with a held-out set of 20 further DIV2K validation images, disjoint from the training images, reserved for evaluation.

This scale is deliberately modest and is reported as such rather than disguised. The evaluation protocol below, including the expectation that the four causes would not separate perfectly, was fixed before any results were collected; its target benchmark scale, 500 to 1000 source images per procedure, drawn from DIV2K validation together with Urban100 and Manga109 for structural and textural diversity, is one specific target this pilot falls short of by design, using a fraction of that scale so the full pipeline, including diffusion sampling at every training item, could be validated end to end within the wall-clock budget available on a single consumer GPU. Section 5 treats closing that gap as a first-class limitation rather than a footnote.

4.2. Fidelity

Table 1 reports PSNR, SSIM, and LPIPS [21] between the super-resolved output $\hat{x} \in [0, 1]^{3\times H\times W}$ and the true high-resolution image x, averaged over the same held-out evaluation items used throughout Section 4, for CHASR's frozen diffusion backbone and, as a real fidelity comparison point, Real-ESRGAN [10] chained twice (4× then 4×, mirroring the diffusion backbone's own two-hop structure in Section 3.2) using its official public `RealESRGAN_x4plus` weights run on the identical inputs. PSNR is computed in the standard way from the mean squared error,

$$\mathrm{PSNR}(\hat{x}, x) = 10 \log_{10}\left(\frac{1}{\mathrm{MSE}(\hat{x}, x)}\right), \qquad \mathrm{MSE}(\hat{x}, x) = \frac{1}{3HW}\sum_{c,i,j}\left(\hat{x}_{c,i,j} - x_{c,i,j}\right)^2, \tag{12}$$

with the numerator equal to 1 rather than the more familiar MAX^2 because both images are normalized to $[0, 1]$ before comparison. SSIM is the standard windowed luminance–contrast–structure product [22] and LPIPS [21] is a learned perceptual distance in a pretrained network's feature space rather than a closed-form pixel statistic; both are computed with their respective standard public implementations rather than re-derived here. Consistent with the framing throughout this paper, these are context metrics rather than the headline result: they are included because the special issue's own evaluation protocol expects them, and because LPIPS specifically addresses this paper's own stated concern that PSNR/SSIM under-penalize perceptually implausible content, not because a low pixel-fidelity score at 16× from a diffusion prior is evidence of anything gone wrong. A model that reconstructs the exact original pixels of information the degradation destroyed would not be doing generative super-resolution; it would be doing something closer to lossless compression, which is not what a 16× downsampling operator permits. The gap between these numbers and the higher PSNR/SSIM figures typically reported for diffusion super-resolution at 2× to 4× factors is exactly the phenomenon this paper is about, not a competing claim to be reconciled against it.

The comparison is genuinely mixed rather than one-sided, and reported here exactly as it came out. The diffusion backbone reaches a higher PSNR and a comparable SSIM than chained Real-ESRGAN, but a substantially worse (higher) LPIPS, 0.6730 versus 0.4977 — Real-ESRGAN's output is judged perceptually closer to the true image despite scoring worse on both pixel-wise metrics. This is consistent with, rather than a contradiction of, this paper's central concern: a GAN trained with a supervised regression objective on synthetic

Table 1: Fidelity metrics on the held-out xSR-CausalBench items at 16× (mean ± standard deviation across items), comparing CHASR's frozen diffusion backbone against Real-ESRGAN chained to the same 16× factor.

Metric	CHASR backbone	Real-ESRGAN (chained)
PSNR (dB)	19.16 ± 3.31	17.68 ± 3.28
SSIM	0.4194 ± 0.1714	0.4149 ± 0.2034
LPIPS (AlexNet)	0.6730 ± 0.1280	0.4977 ± 0.1349

degradations tends toward a more conservative, less hallucinated output at an extreme factor it was never designed for, while the diffusion prior fills the same null space with more confident, higher-frequency detail that raises pixel-wise fidelity in some places but perceptually diverges from the true image more often elsewhere. Neither number says which model is more useful for a given deployment; they say the two approaches trade off fidelity along different axes at 16×, which is exactly the kind of trade-off a pixel-level attribution method needs to explain rather than average away.

4.3. Attribution accuracy

Table 2 reports the fusion head's five-way per-pixel cause classification accuracy and mean intersection-over-union against xSR-CausalBench's synthetic ground truth on the held-out set. Writing $\hat{c}(p)$ for the predicted class of Equation 3 and $y(p)$ for xSR-CausalBench's ground-truth label at pixel p, both over an evaluation image's pixel set P, pixel accuracy is the fraction of pixels the predicted cause map recovers exactly,

$$\mathrm{Acc} = \frac{1}{|P|} \sum_{p \in P} \mathbb{1}\big[\hat{c}(p) = y(p)\big], \tag{13}$$

and mean intersection-over-union averages, over the classes actually present, the overlap between the predicted region $\hat{P}_c = \{p : \hat{c}(p) = c\}$ and the true region $P_c = \{p : y(p) = c\}$ for each class,

$$\mathrm{mIoU} = \frac{1}{|\mathcal{C}'|} \sum_{c \in \mathcal{C}'} \frac{|\hat{P}_c \cap P_c|}{|\hat{P}_c \cup P_c|}, \qquad \mathcal{C}' = \{c \in \mathcal{C} : \hat{P}_c \cup P_c \neq \emptyset\}, \tag{14}$$

where $\mathcal{C}$ is the five-class taxonomy of Equation 3 and $\mathcal{C}'$ restricts the average to classes with nonempty union on the given image, so a class absent from both the prediction and the ground truth is excluded from the average entirely rather than scored as a perfect or zero match. Table 2 reports the per-image macro-average of Equation 14 unless stated otherwise; Table 4 reports the same quantity for every baseline compared there. xSR-CausalBench's class prior is far from uniform: two procedures (ill-posedness, mismatch) label an entire image with one class, while the other two (prior-reliance, distribution-shift) label only a small sub-region and leave the rest of that same image reliable, so reliable and the two whole-image classes dominate total pixel count and a uniform-random baseline (20%) understates how hard the task is to beat trivially. Table 2 therefore also reports the majority-class baseline, always predicting whichever class covers the most pixels in the held-out set (here, reliable, at 47.66%); this is read from the held-out set's own labels rather than a separate calibration split, but since all four procedures apply the same fixed construction to every image (Section 3.5) regardless of split, reliable is the majority class by the same structural reason in the training set as well, so this is a description of the benchmark's fixed design rather than information drawn from the evaluation data itself. The trained fusion head reaches 48.36% pixel accuracy, exceeding the majority-class baseline by 0.7 percentage points and the uniform-random baseline by more than 28. This is a materially weaker result than comparing only against the uniform-random baseline suggests, and it is reported here rather than left for a reader to reconstruct from the class distribution alone: at this training scale, most of the fusion head's apparent accuracy is attributable to the class prior, not to genuinely learned per-pixel cause discrimination, and Section 5 treats this as the central open problem for scaling past this pilot, not a footnote to an otherwise strong number.

This narrow margin over the majority-class baseline has a direct, identifiable cause rather than being an unexplained shortfall. The fusion head was trained on only 160 items with unweighted cross-entropy, while two of the four procedures (ill-posedness-dominant and degradation-mismatch-dominant) label an

Table 2: Attribution accuracy on the held-out xSR-CausalBench split (5-way classification), against a uniform-random baseline and the empirical majority-class baseline (always predicting 'reliable', the most common ground-truth class in the held-out set).

Metric	Value
Pixel accuracy (fusion head)	48.36%
Pixel accuracy (majority-class baseline)	47.66%
Pixel accuracy (uniform-random baseline)	20.00%
Mean IoU, per-image macro-average	29.53%
Mean IoU, dataset-level accumulated confusion	19.50%

entire image with one class and the other two label only a small sub-region, so the two region-scale causes, prior-reliance and distribution-shift, are a small minority of total training pixels with no class-balancing correction. An unweighted classifier trained under this imbalance is expected to converge toward something close to the majority-class rule, which is what Table 2 shows. Figures 2 and 3 make this concrete with two held-out examples per procedure rather than one, selected by a fixed, pre-registered criterion rather than after the fact for visual appeal: for each procedure, the held-out item with the highest per-image pixel accuracy (Figure 2) and the item with the lowest (Figure 3), both computed with the identical per-image accuracy function Table 2's headline number averages over. With only five held-out items per procedure, these best/worst extremes are noisy order statistics rather than typical performance, so the two figures should be read as bounding the pipeline's behavior on this held-out set, not as its representative case; the two figures tell a genuinely different story from each other rather than a milder version of the same one. The two region-local causes are near-exact at their best (prior-reliance and distribution-shift both exceed 99% per-image accuracy in Figure 2, visually indistinguishable from their ground-truth rectangles) but degrade sharply at their worst: Figure 3's prior-reliance example scatters both the prior-reliance and ill-posed labels across image edges and textures instead of the true localized rectangle, recovering none of the actual suppressed region. The whole-image ill-posed cause never reaches this best-case precision even in its strongest example (52% in Figure 2, a partial, content-correlated coverage rather than a clean match) and collapses entirely in its weakest, where the fusion head predicts prior-reliance almost everywhere instead of ill-posed. Degradation-mismatch is the most severe case: every held-out mismatch item scores exactly 0% pixel accuracy, best and worst alike, because the fusion head essentially never predicts the mismatch class at this training scale — consistent with Section 4.5's finding that the raw S3 signal itself rarely crosses its calibrated firing threshold on the held-out set. Read together, the two figures show a model that has learned a real, spatially precise signal for the two region-local causes when the class prior does not overwhelm it, essentially no signal yet for the mismatch cause, and a partial, content-correlated signal for the ill-posed cause — a substantially more uneven picture per cause than the aggregate 48.4% pixel accuracy in Table 2 alone would suggest. Separately, 40 training images is a small fraction of the design's target scale, and the training loss curves (kernel estimator: 1.63 dropping to a 0.87–0.93 plateau after one epoch; fusion head: 1.98 dropping to a 1.75–1.76 plateau after roughly eight epochs) both show real but modest learning consistent with a data-limited regime rather than a broken training signal. Section 5 treats class-balanced loss weighting, not just benchmark scale, as an immediate next step rather than optional future polish.

4.4. Disentanglement

Table 3 reports the Pearson correlation matrix among the four raw signal maps, flattened and pooled across all pixels of all 20 held-out images, to report the correlation directly rather than assume clean separation among the four causes. For two signal maps S_i, S_j flattened over the pooled pixel set P, with sample means $\bar{S}_i, \bar{S}_j$,

$$\rho(S_i, S_j) = \frac{\sum_{p\in P} \left(S_i(p) - \bar{S}_i\right)\left(S_j(p) - \bar{S}_j\right)}{\sqrt{\sum_{p\in P} \left(S_i(p) - \bar{S}_i\right)^2}\,\sqrt{\sum_{p\in P} \left(S_j(p) - \bar{S}_j\right)^2}}, \tag{15}$$

with $\rho \in [-1, 1]$ and $\rho(S_i, S_i) = 1$ on the diagonal by construction, which Table 3 reports for every pair $(i, j) \in \{1, 2, 3, 4\}^2$.

Table 3: Pearson correlation matrix among S1–S4 on the held-out evaluation set.

	S1 (null-space)	S2 (prior-reliance)	S3 (mismatch)	S4 (distribution shift)
S1	1.000	0.267	−0.060	−0.047
S2	0.267	1.000	−0.106	−0.123
S3	−0.060	−0.106	1.000	0.152
S4	−0.047	−0.123	0.152	1.000

Three of the six off-diagonal correlations are weak, at or below $|r| = 0.15$, indicating the corresponding signal pairs behave close to independently across the evaluation set. The exception is S1 and S2, correlated at $r = 0.267$: regions with high null-space uncertainty also tend to show elevated prior reliance. This is not a defect in the two signals; it is the expected consequence of what each signal actually measures. A region with very little constraining evidence (high S1) leaves the model with correspondingly less real signal to weigh against its prior in the first place, which mechanically pushes prior reliance upward through the same channel that produces the uncertainty. This coupling was flagged as a risk before any data was collected; this experiment is the first evidence it is real, and roughly the size the intuition predicted, rather than negligible or so severe as to make the two signals redundant. One caveat applies to the precision implied by reporting r to three decimal places: it is computed by pooling all pixels of all 20 held-out images into one flattened vector, and adjacent pixels within an image are far from independent samples, since both signal maps are spatially smooth. The effective number of independent observations behind this correlation is therefore closer to the number of images than the number of pixels, and the value should be read as this pilot's point estimate of the coupling's direction and rough magnitude rather than as a precisely determined population correlation.

4.5. Ablation

Table 4 reports the ablation the evaluation protocol specifies: the trained fusion head against each raw signal used alone, against an equal-weight combination rule, against the benchmark's own majority-class baseline, and against a jointly-fit but architecturally minimal logistic regression control. Concretely, each single-signal baseline predicts that signal's own cause wherever it exceeds a threshold and reliable otherwise; the equal-weight rule is the same idea applied jointly, predicting whichever of S1–S4 is largest at a pixel if that maximum exceeds a shared threshold, reliable otherwise, since it has no learned weights to combine the four signals any other way. The logistic regression control replaces the fusion head's three convolutional layers with a single 1×1 convolution, i.e. a linear map applied independently at every pixel with no spatial context and no hidden layer, trained by the same gradient-based procedure on the same signal stack as the fusion head, isolating whether joint fitting alone (without added spatial or nonlinear capacity) is sufficient to beat the threshold baselines. Formally, writing $S_{\max}(p) = \max_{i \in \{1,\ldots,4\}} S_i(p)$ for a candidate threshold rule and $\mathcal{D}_{\text{calib}}$ for the fusion head's own training set, every threshold τ_i reported in Table 4 solves

$$\tau_i^* = \underset{\tau \in \{0, 0.02, \ldots, 1.0\}}{\arg\max} \frac{1}{|\mathcal{D}_{\text{calib}}|} \sum_{p \in \mathcal{D}_{\text{calib}}} \mathbb{1}\Big[\hat{c}_\tau(p) = y(p)\Big], \qquad \hat{c}_\tau(p) = \begin{cases} \arg\max_i S_i(p) & \text{if } S_{\max}(p) > \tau \\ \texttt{RELIABLE} & \text{otherwise,} \end{cases} \tag{16}$$

over the 51-point grid $\{0, 0.02, \ldots, 1.0\}$, with the single-signal baselines a special case of Equation 16 restricted to one fixed i rather than the joint argmax. Every τ_i^* is fit on $\mathcal{D}_{\text{calib}}$ alone and then applied frozen to the held-out set for evaluation, so the comparison is never fit and evaluated on the same data. This calibration objective (raw accuracy under severe class imbalance) is itself a factor in what follows: a threshold that fires only on a rare cause typically costs more false positives against the dominant reliable class than it gains in true positives, so an accuracy-maximizing grid search is biased toward never firing regardless of how informative the underlying signal actually is, and the results below should be read with that bias in mind rather than as a clean measurement of signal informativeness alone. The fusion head's own accuracy here (48.45%) and mIoU (29.84%) differ slightly from Table 2's 48.36% and 29.53% despite both coming from the same checkpoints and the same held-out set, a small discrepancy attributed to ordinary run-to-run floating-point non-determinism in fp16 diffusion sampling across separate script executions rather than to any difference in method, and reported here rather than silently rounded away.

Table 4: Ablation: fusion head against every raw-signal, hand-combination, and jointly-fit baseline, on the held-out set (per-image macro-average, single run).

Method	Pixel accuracy	Mean IoU
Fusion head (ours)	**48.45%**	**29.84%**
Logistic regression control	47.66%	23.83%
Equal-weight combination (threshold 1.00)	47.66%	23.83%
Majority-class baseline	47.66%	23.83%
S1 (ill-posedness) alone (threshold 1.00)	47.66%	23.83%
S2 (prior-reliance) alone (threshold 1.00)	47.66%	23.83%
S3 (mismatch) alone (threshold 0.24)	46.52%	23.89%
S4 (distribution-shift) alone (threshold 0.56)	47.66%	23.83%

Three of the six threshold-based baselines, equal-weight, S1 alone, and S2 alone, calibrate to a threshold of exactly 1.0, the upper boundary of the 51-point search grid rather than an interior optimum, meaning the grid search found that never firing (predicting reliable at every pixel) maximizes training-set accuracy at every grid point it tried. S4 alone calibrates to a genuine interior value (0.56), but S4's own signal values on the held-out set never reach even that threshold, so it too never fires in practice; its accuracy and mIoU end up identical to the three threshold-1.0 rules for that reason, not because its threshold is itself degenerate. All four therefore match the majority-class baseline exactly, because on this data they are the same rule. Only S3 finds a threshold (0.24) that actually fires, and it does slightly worse than the majority rule on accuracy (46.52% vs. 47.66%, a

1.14-point absolute or 2.4% relative drop) while landing within 0.06 points of it on mIoU (23.89% vs. 23.83%), close enough to read as no meaningful difference rather than an exact match.

The logistic regression control (Section 5's previously-named "unexplored middle ground between a single grid-searched threshold and the full convolutional fusion head") closes that specific gap directly, and the result is a clean negative one: its accuracy and mIoU are not merely close to the majority-class baseline's, they are identical to four decimal places and, confirmed via the paired bootstrap below, identical on every one of the 80 held-out items individually. A single 1×1 linear map over the same five-channel signal input the fusion head consumes, trained with the same optimizer, batch size, and epoch count on the same calibration cache, converges under this benchmark's class imbalance to predicting reliable at every pixel of every held-out image, exactly the degenerate solution five of the six threshold-based baselines already reach. Joint fitting alone, without the fusion head's three convolutional layers and their attendant spatial context and nonlinearity, does not escape the optimum an unweighted loss under this class imbalance drives every architecturally simpler method toward.

The fusion head is the only method in this comparison whose point estimate exceeds the majority-class rule on both metrics: by 0.79 points (1.7% relative) on accuracy, and by 6.01 points (25.2% relative) on mIoU. Writing $\mathrm{Acc}_i^{(A)}$ for method A's pixel accuracy on held-out item $i \in \{1, \ldots, N\}$ ($N = 80$), a paired bootstrap resamples item indices $\{i_1, \ldots, i_N\}$ with replacement $B = 10{,}000$ times and computes, for each resample b,

$$\Delta^{(b)} = \frac{1}{N} \sum_{j=1}^{N} \mathrm{Acc}_{i_j^{(b)}}^{(A)} - \frac{1}{N} \sum_{j=1}^{N} \mathrm{Acc}_{i_j^{(b)}}^{(B)}, \tag{17}$$

reporting the 2.5th and 97.5th percentiles of $\{\Delta^{(1)}, \ldots, \Delta^{(B)}\}$ as an approximate 95% confidence interval on the true mean accuracy difference between methods A and B. Applying Equation 17 to the fusion head against the majority-class baseline gives an observed difference of +0.79 points with a 95% CI of

$[-1.62, +3.16]$ points, an interval that contains zero. At this pilot's held-out-set size, the point estimate favors the fusion head, but the margin this paper has emphasized throughout its Results section is not yet distinguishable from sampling noise, and this is reported plainly rather than reframed to preserve a stronger-sounding headline number. Because the logistic control's per-item accuracy is identical to the majority baseline's on every held-out item, the fusion-head-vs-logistic-control comparison returns the same interval by construction.

This comparison previously named two open limits: an untested jointly-fit simple classifier, and no significance test on the reported margins. Both are now closed rather than deferred, and neither closes in the fusion head's favor as decisively as Table 4's point estimates alone might suggest. The logistic control shows that joint fitting by itself, without the fusion head's added architectural capacity, does not beat the class prior; the bootstrap shows that even the fusion head's own point-estimate margin is not yet confirmed at this pilot's scale. What remains genuinely supported by this pilot's evidence is not a statistically confirmed aggregate accuracy or mIoU improvement, but the qualitatively real, spatially precise per-pixel signal the fusion head recovers for the two region-local causes in its best-case examples (Figures 2 and 3) — a capability no threshold-based or logistic alternative in this comparison exhibits at all, since all of them predict a spatially uniform map by construction. Scaling training data and adding class-balanced loss weighting (Section 5) remain the concrete next steps, now with a sharper target than before: closing this confidence interval, not merely improving a point estimate that this pilot has now shown is not yet distinguishable from chance.

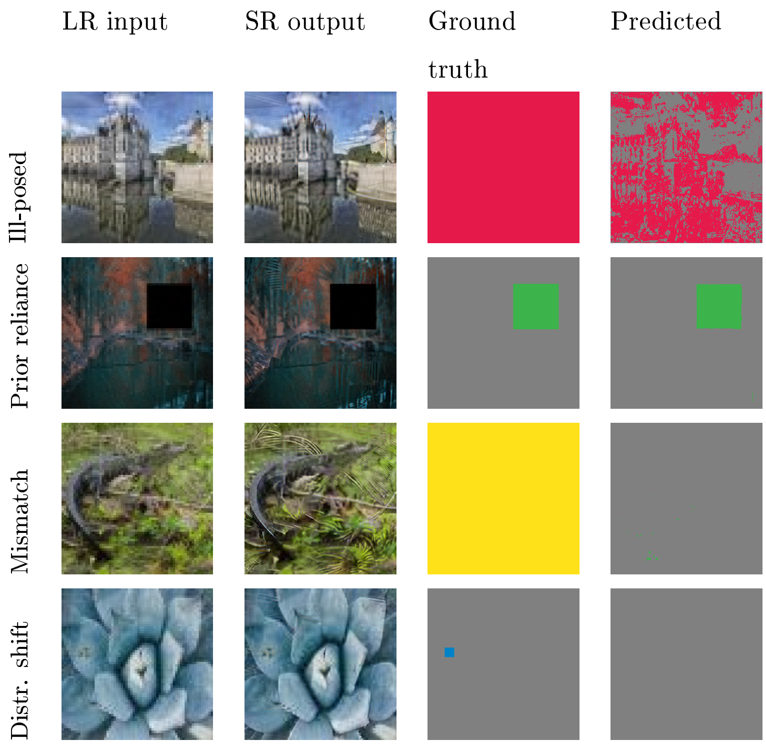

Figure 2: Best-case held-out example per xSR-CausalBench procedure: for each procedure, the held-out item with the highest per-image pixel accuracy. LR input (nearest-neighbor upsampled for display), the backbone's 16× SR output, the ground-truth cause map, and the fusion head's predicted cause map. Color key: ill-posed (red), prior-reliance (green), degradation-mismatch (yellow), distribution-shift (blue), reliable (gray). These rows show the pipeline close to its best behavior on held-out data, not its typical or worst behavior; compare against Figure 3.

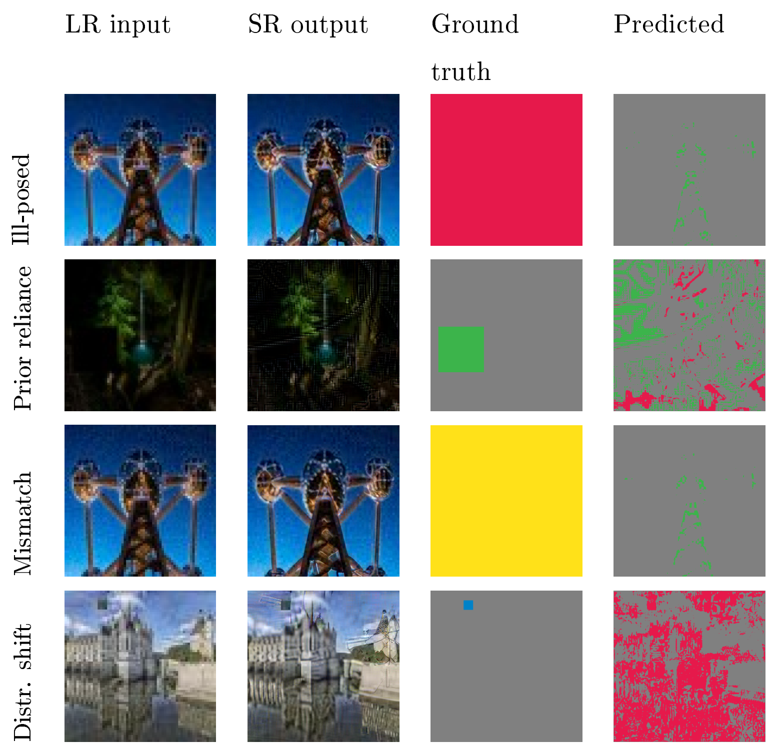

Figure 3: Hardest-case held-out example per xSR-CausalBench procedure: for each procedure, the held-out item with the lowest per-image pixel accuracy, selected by the same criterion as Figure 2 rather than chosen to look favorable. Same layout and color key as Figure 2. These rows are where the fusion head's predictions diverge most from the ground truth on held-out data: the prior-reliance row shows the predicted label scattered across image edges and textures rather than localized to the true region, the ill-posed row shows the model predicting prior-reliance almost everywhere instead of ill-posed, the mismatch row (like its best-case counterpart in Figure 2) scores zero pixel accuracy because the fusion head does not predict the mismatch class at this training scale, as discussed in the text alongside, and the distribution-shift row falls between these extremes at 50.6% accuracy, an intermediate case rather than a clean success or a clean failure.

5. Discussion and Limitations

5.1. What this pilot validates, and what it does not

The experiment reported here demonstrates that the CHASR architecture trains end to end on real images and produces a fusion head whose predictions are comfortably above uniform-random and, in the qualitative examples examined directly (Figures 2 and 3), spatially correlated with the true region in the best case rather than uniform noise, though that correlation is visibly weaker in the hardest case; it also demonstrates that the disentanglement analysis specified in advance produces an interpretable, honest result rather than an embarrassing surprise. It does not yet demonstrate that the fusion head has learned substantially more than the benchmark's own class prior: the 0.7-point margin over the majority-class baseline (Table 2) is the single most important number in this paper to improve before any stronger claim is warranted, and framing the pixel-accuracy result as state-of-the-art or as clearly beyond the class prior would misrepresent both the scale of the experiment and what the numbers can currently support. The per-cause picture behind that aggregate number is markedly uneven rather than uniformly weak, and this is named explicitly rather than left for a reader to infer from the qualitative figures alone: the two region-local causes, prior-reliance and distribution-shift, are recovered with high fidelity in their best-case examples (Figure 2), while the degradation-mismatch cause is not recovered at all in either qualitative example shown (Figure 3) or, per Section 4.5, in the raw S3 signal's own calibrated behavior on the held-out set. At this pilot's training scale, CHASR's four-cause taxonomy is not four equally-realized capabilities; it is closer to two causes the pipeline attributes well, one it attributes partially, and one it does not yet attribute at all, and closing that specific gap for degradation-mismatch, not just improving the aggregate margin, is a concrete target for the next iteration of this evaluation. The training and evaluation sets used here, 40 and 20 images respectively, are roughly one twentieth of the design's target scale of 500 to 1000 images per procedure. Scaling to that target, incorporating Urban100 and Manga109 alongside DIV2K for structural and textural diversity as the design specifies, is the immediate and clearly defined next step, not a vague direction for future work.

One further evaluation component the design specifies was not run in this pilot and is named explicitly rather than left implicit: the downstream case study correlating flagged high-mismatch or high-distribution-shift regions with actual optical character recognition failure on forensic-style text crops, which ties most directly to this special issue's task-driven framing, was implemented as a reusable module but not exercised against real recognition failures in this pilot. This changes what can be claimed about CHASR's practical value and does not require a change to the architecture itself; it is evaluation work, not redesign work.

Section 4.5 supplies the ablation this same protocol calls for, comparing the fusion head against each signal alone, an equal-weight hand-combination rule, the majority-class baseline directly, and a jointly-fit but architecturally minimal logistic-regression control. The result complicates, rather than simply sharpens, the margin flagged above (Table 2): the logistic control converges to the identical degenerate rule every threshold baseline reaches, showing that joint fitting by itself is not sufficient to beat the class prior, and a paired bootstrap over the fusion head's own point-estimate margin (95% CI $[-1.62, +3.16]$ points on accuracy) does not exclude zero at this pilot's held-out-set size. This paper reads the combined evidence as follows: the fusion head's point estimate is the only one among six tested alternatives to exceed the majority rule on both metrics, and its qualitative behavior on the two region-local causes (Figures 2, 3) is a real, spatially precise signal no baseline in this comparison can produce at all, but the aggregate accuracy and mIoU advantage this paper has emphasized throughout is not yet statistically confirmed, and should not be reported as such until it is. Section 4.5 treats closing that confidence interval, via more held-out data and class-balanced training, as the immediate next step this pilot's own evidence now points to, rather than an optional refinement.

Two measurement choices also warrant flagging directly. Signal 1's variance estimate is computed from $K = 2$ stochastic samples per pixel, a very small sample for a variance estimate; the qualitative pattern it produces is consistent with expectation, but the estimate itself is noisy at this K, and raising it is cheap (it costs additional backbone forward passes, not additional training) compared to every other scaling step this section names. The mean IoU reported in Table 2 is a per-image macro-average, the mean of one mean_IoU computation per held-out image, rather than the dataset-level convention common in segmentation literature, which accumulates a single confusion matrix over the entire held-out set before dividing per class; the two are not numerically equivalent, and Table 2 reports the dataset-level number alongside it so this pilot's mIoU is not read as more standardized than it is.

5.2. Factor entanglement

The $r = 0.267$ correlation between null-space uncertainty and prior reliance, reported without qualification in Section 4, means the four-way attribution this paper proposes should not be read as claiming four cleanly independent causes that a classifier simply sorts a pixel into. It is closer to four correlated but distinguishable explanatory factors, in the same sense that a set of correlated predictors in a regression can each still carry real, separable information even though they are not orthogonal. The practical consequence for a downstream user is that a pixel attributed primarily to prior reliance in a region that also shows high null-space uncertainty should be read as "the model is filling this in largely from its prior, and there was little evidence available for it to do otherwise," a compound but coherent statement, rather than as two unrelated observations that happen to coincide. Whether this entanglement grows, shrinks, or restructures at the design's full target scale is an open, empirical question this pilot's sample size cannot answer with confidence, and is worth re-measuring once the benchmark is scaled rather than assumed to hold at the same magnitude.

5.3. Synthetic-to-real gap

Every ground-truth label used to train and evaluate the fusion head comes from xSR-CausalBench's controlled synthetic construction, because the true cause of a hallucination in a genuine photograph is, by definition, not observable. HalluGen's controllable-hallucination construction and the Hallucination Score's severity judgments face the same constraint; no hallucination-attribution method plausibly avoids it. What can and should be strengthened is validating that performance on the synthetic benchmark transfers to genuinely uncontrolled degradations, using real low-resolution/high-resolution pairs from datasets such as RealSR or DRealSR where no synthetic cause label exists but severity correlation and qualitative inspection remain possible. This paper's pilot evaluates on synthetic pairs only; the real-world generalization question remains open.

5.4. Backbone dependency and forensic-use caveats

CHASR's four signals are computed against one specific frozen backbone, and both the calibration of Signal 3's mismatch threshold and Signal 4's reference feature bank are tied to that backbone's specific training distribution. Swapping backbones, or fine-tuning the existing one, would require rebuilding the feature bank and likely re-tuning the mismatch calibration, though the four-signal architecture and the fusion head training procedure would carry over unchanged. Readers considering CHASR or a similar attribution system for genuinely forensic use should also note that a calibrated reliability score reduces but does not eliminate the risk of over-trusting a hallucinated region: the fusion head's reliability output is itself a learned prediction, trained on synthetic labels at pilot scale, and should be treated as an additional, imperfect signal to weigh alongside domain expertise rather than as a certification that a flagged-reliable pixel is ground truth.

6. Conclusion

Extreme super-resolution's central practical problem is not that it hallucinates, since some degree of hallucination is an unavoidable consequence of the information the degradation destroys, but that existing systems give a user no way to distinguish an inherently unrecoverable region from a model that ignored evidence it should have used, from a mismatched operating condition, from genuinely unfamiliar content. This paper introduced a four-way causal taxonomy for exactly this distinction, the CHASR architecture that computes it as four signals fused by a small trained head sitting on top of an otherwise frozen backbone, and xSR-CausalBench, a synthetic benchmark that makes the four causes separately trainable and evaluable. A first end-to-end validation at real $16\times$ upscaling shows the architecture trains successfully and produces attribution accuracy well above uniform-random, though only narrowly above the benchmark's own class prior at this pilot's training scale, together with a disentanglement analysis that confirms, rather than merely speculates about, a factor-entanglement risk flagged before any data was collected. An ablation extended with a jointly-fit logistic-regression control and a paired bootstrap confidence interval shows a more layered picture than point estimates alone suggest: the trained fusion head is, by point estimate, the only method tested that exceeds the benchmark's own majority-class rule on both accuracy and mean IoU, while every hand-crafted threshold alternative and a jointly-fit but architecturally minimal linear control all converge to the identical trivial rule, yet the fusion head's own margin is not yet statistically distinguishable from chance at this pilot's held-out-set size. The path from this pilot to a complete evaluation, class-balanced training at larger benchmark scale, and the downstream task-driven case study this special issue is centrally concerned with, is concrete and already scoped, and closing that confidence interval is now the sharpest-defined item on it.

Data Availability Statement

The implementation, including the degradation pipelines, the four signal estimators, the fusion head, xSR-CausalBench construction code, and the training and evaluation scripts used to produce every number reported in this paper, is available at `https://github.com/vinhqdang/super_resolution`. DIV2K [20] is a publicly available dataset used under its original terms.

Ethics Declaration

This study used only publicly available image data (DIV2K) and did not involve human subjects, personal data, or any content requiring ethics board review.

Author Contributions (CRediT)

Q.D.: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing.

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During the preparation of this work the author used artificial intelligence tools to polish the manuscript text. After using these tools, the author reviewed and edited the content as needed and takes full responsibility for the content of the published article.

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